

Lesson 7 - Your First Models

READ ME FIRST (theory). Then open the practice notebook.

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Lesson 7 - Your First Models

Now we add a model. The theory, then `practice.ipynb`.

Key words

Word	Simple meaning
model	the thing that learns the pattern and predicts
Linear Regression	predicts a number
Logistic Regression	predicts a category (Yes/No) with a probability
fit	train the model on data
predict	get answers for new data

1. What is a model?

A model learns the pattern between the **features** (inputs) and the **label** (answer). After training, it can answer for new data it has not seen.

2. Linear Regression - predict a number

Linear Regression draws the best straight-line formula to predict a **number** (price, sales).

Use it when: the answer is a number.

3. Logistic Regression - predict Yes/No

Logistic Regression gives a **probability** between 0 and 1, so we can answer a **category** like churn Yes/No. If probability > 0.5 -> Yes.

Use it when: the answer is a category (especially two categories).

Common mistake: using Linear Regression for Yes/No. Use Logistic for categories.

4. Two steps: fit then predict

```
model.fit(X_train, y_train)    # STEP 1: learn from training data
y_pred = model.predict(X_test) # STEP 2: answer for new data
```

Common mistake: calling predict before fit. The model must be trained first, or it gives an error.

5. Put the model at the end of the Pipeline

We add the model as the last step of the Pipeline from Lesson 6. Now ONE object does everything: fill -> scale -> encode -> predict.

```
from sklearn.linear_model import LogisticRegression
model = Pipeline([
    ("preprocessor", preprocessor),
    ("clf", LogisticRegression(max_iter=1000)),
])
model.fit(X_train, y_train)
```

Quick summary

- A **model** learns features -> label.
- **Linear Regression** predicts a number. **Logistic Regression** predicts a category.
- Always **fit** (train) before **predict**.
- Put the model as the last step of the Pipeline = one object does everything.

Now do this

Open `practice.ipynb`, train a Logistic Regression model, and do the assignment.